

Problem 4.8.26

Sarvesh Tamgade

September 17, 2025

Question

Question: Find the coordinates of the foot of the perpendicular drawn from the point

$$\mathbf{P} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$$

to the Y -axis.

Solution

The Y -axis has the direction vector

$$\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

and passes through the origin. Its general point is

$$\mathbf{Q} = \begin{pmatrix} 0 \\ q \\ 0 \end{pmatrix}.$$

Any point $\mathbf{Q}$ on the Y -axis satisfies $x = 0$ and $z = 0$.

Let $\mathbf{P} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}.$

The foot of the perpendicular $\mathbf{Q}$ is given by projecting $\mathbf{P}$ onto the Y -axis as

$$\mathbf{Q} = (\mathbf{e}_2^\top \mathbf{P}) \frac{\mathbf{e}_2}{\|\mathbf{e}_2\|^2}.$$

Solution

$$\mathbf{e}_2^\top \mathbf{P} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}.$$

Since

$$\|\mathbf{e}_2\|^2 = 0^2 + 1^2 + 0^2 = 1,$$

the foot of the perpendicular is

$$\mathbf{Q} = \left(\begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \right) \mathbf{e}_2.$$

$$\mathbf{Q} = (-3)\mathbf{e}_2 = \begin{pmatrix} 0 \\ -3 \\ 0 \end{pmatrix}.$$

Final Answer: The coordinates of the foot of the perpendicular are

$$\begin{pmatrix} 0 \\ -3 \\ 0 \end{pmatrix}.$$

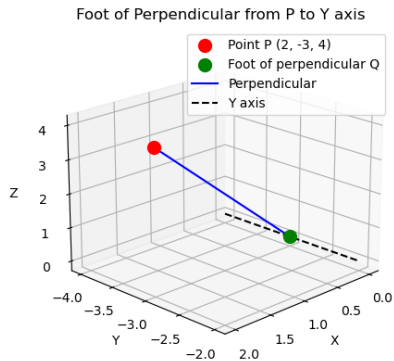


Figure: Vector Representation

```
#include "matfun.h"

// Calculate dot product of two 3D vectors
double dot_product(const double a[3], const double b[3]) {
    return a[0]*b[0] + a[1]*b[1] + a[2]*b[2];
}

// Compute the foot of the perpendicular from point P to the Y-axis
void foot_of_perpendicular_to_Y_axis(const double P[3], double Q[3]) {
    // Direction vector of Y-axis
    double m[3] = {0.0, 1.0, 0.0};
    double m_norm_sq = dot_product(m, m);

    // dot product of m and P
    double dp = dot_product(m, P);
```

```
// projection scalar
double scalar = dp / m_norm_sq;

// Q = scalar * m
Q[0] = scalar * m[0];
Q[1] = scalar * m[1];
Q[2] = scalar * m[2];
}
```


Python Code - Using Shared Object

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Load the shared library
matfun_lib = ctypes.CDLL("./matfun.so")

# Define argument types for the C function
# void foot_of_perpendicular_to_Y_axis(const double P[3], double
    Q[3])
matfun_lib.foot_of_perpendicular_to_Y_axis.argtypes = [
    np.ctypeslib.ndpointer(dtype=np.double, ndim=1, flags="
        C_CONTIGUOUS"),
    np.ctypeslib.ndpointer(dtype=np.double, ndim=1, flags="
        C_CONTIGUOUS")
]
```

Python Code - Using Shared Object

```
# Input point P
P = np.array([2.0, -3.0, 4.0], dtype=np.double)
Q = np.zeros(3, dtype=np.double) # Output array

# Call the C function to compute the foot of perpendicular
matfun_lib.foot_of_perpendicular_to_Y_axis(P, Q)

# Print the result as main.c does
print(f"Foot of the perpendicular from P(2, -3, 4) to Y-axis is:
      ({Q[0]:.2f}, {Q[1]:.2f}, {Q[2]:.2f})")

# Y-axis vector for plotting
y_axis = np.array([[0, 0], [min(P[1], Q[1]) - 1, max(P[1], Q[1])
      + 1], [0, 0]])

# Plotting
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```

Python Code - Using Shared Object

```
ax.scatter(P[0], P[1], P[2], color='r', s=100, label='Point P (2,  
-3, 4)')  
ax.scatter(Q[0], Q[1], Q[2], color='g', s=100, label='Foot of  
perpendicular Q')  
ax.plot([P[0], Q[0]], [P[1], Q[1]], [P[2], Q[2]], color='b',  
label='Perpendicular')  
ax.plot(y_axis[0], y_axis[1], y_axis[2], color='k', linestyle='--  
, label='Y axis')  
  
ax.set_xlabel('X')  
ax.set_ylabel('Y')  
ax.set_zlabel('Z')  
ax.set_title('Foot of Perpendicular from P to Y axis')  
ax.legend()  
ax.view_init(elev=20, azim=45)  
  
plt.savefig('foot_of_perpendicular_from_c.png')  
plt.show()
```